

Matthew D. Kvalheim

Research Interests

My research draws on differential topology and other geometric-topological methods to study nonlinear systems. In particular, I seek to determine when dynamical systems can be stabilized using feedback control or represented by linear or lower-dimensional models.

Education

- 2013–2018 **Ph.D.**, *University of Michigan*, Electrical Engineering
- 2013–2017 **M.S.**, *University of Michigan*, Mathematics
- 2013–2017 **M.S.**, *University of Michigan*, Electrical Engineering
- 2009–2013 **B.S.**, *Ohio University*, Electrical Engineering

Employment

- 2023– **Assistant Professor of Mathematics**, *University of Maryland, Baltimore County*, Department of Mathematics and Statistics
- 2022–2023 **Postdoctoral Assistant Professor**, *University of Michigan*, Department of Mathematics
Mentor: Anthony M. Bloch
- 2019–2022 **Postdoctoral Researcher**, *University of Pennsylvania*, Department of Electrical and Systems Engineering
Mentors: Yuliy Baryshnikov, Daniel E. Koditschek
- 2018–2019 **Postdoctoral Research Fellow**, *University of Michigan*, Department of Electrical Engineering and Computer Science
Mentors: Anthony M. Bloch, Shai Revzen

Awards and Honors

- 2026 **Young Investigator Program (YIP) Award**, *Air Force Office of Scientific Research*
- 2026 **Early-Career Faculty Excellence Award**, *University of Maryland, Baltimore County*, College of Natural and Mathematical Sciences
- 2025 **Cover article**, *Neural Computation*, Volume 37, Issue 12, MIT Press
- 2013 **Rackham Engineering Award**, *University of Michigan*
- 2013 **Outstanding Senior in Electrical Engineering**, *Ohio University*
- 2013 **First place, EECS Division, Undergraduate Student Expo**, *Ohio University*
- 2012 **Undergraduate Research Award**, *College of Engineering*, *Ohio University*
- 2012 **Outstanding Junior in Electrical Engineering**, *Ohio University*
- 2011 **Dean's Scholarship**, *Ohio University*
- 2011 **Fritz & Dolores Russ Scholarship**, *Ohio University*
- 2009–2013 **Gateway Excellence Scholarship**, *Ohio University*

Funding

- 2026–2029 **A Topological Approach to Simplifying Hybrid Systems**, *Air Force Office of Scientific Research Young Investigator Program (YIP) Award, Grant FA9550-26-1-B628*
Amount: \$448,699.
- 2024–2027 **Geometric-Topological Limitations and Capabilities for Stability and Safety**, *Air Force Office of Scientific Research Grant FA9550-24-1-0299*
Amount: \$260,694.

Peer-Reviewed Journal Publications

- 16 **Linearizability of flows by embeddings**
Matthew D. Kvalheim and Philip Arathoon
Selecta Mathematica, 32.2 (2026), 38. ([article](#), [arXiv](#))
- 15 **Estimating phase from observed trajectories using the temporal 1-form**
Simon Wilshin, Matthew D. Kvalheim, Clayton Scott, and Shai Revzen
Neural Computation, 37.12 (2025), pp. 2158–2204. ([article](#))
- 14 **Global linearization of asymptotically stable systems without hyperbolicity**
Matthew D. Kvalheim and Eduardo D. Sontag
Systems and Control Letters, 203 (2025), 106163. ([article](#), [arXiv](#))
- 13 **Why should autoencoders work?**
Matthew D. Kvalheim and Eduardo D. Sontag
Transactions on Machine Learning Research (2024). ([article](#), [arXiv](#))
- 12 **Phase response curves and the role of coordinates**
Simon Wilshin, Matthew D. Kvalheim, and Shai Revzen
Biological Cybernetics, 118 (2024), pp. 311–330. ([article](#))
- 11 **Flux in tilted potential systems: negative resistance and persistence**
Yuliy Baryshnikov and Matthew D. Kvalheim
Communications in Mathematical Physics, 400.2 (2023), pp. 853–930 ([article](#), [arXiv](#))
- 10 **Obstructions to asymptotic stabilization**
Matthew D. Kvalheim
SIAM J. Control and Optimization, 61.2 (2023), pp. 536–542 ([article](#), [arXiv](#))
- 9 **A compositional approach to certifying the almost global asymptotic stability of cascade systems**
Jake Welde, Matthew D. Kvalheim, and Vijay Kumar
IEEE Control Systems Letters, 7 (2023), pp. 1969–1974. ([article](#), [arXiv](#))
- 8 **A generalization of the Hopf degree theorem**
Matthew D. Kvalheim
Proceedings of the American Mathematical Society, 151.1 (2023), pp. 453–454. ([article](#), [arXiv](#))
- 7 **Necessary conditions for feedback stabilization and safety**
Matthew D. Kvalheim and Daniel E. Koditschek
J. Geometric Mechanics, 14.4 (2022), pp. 659–693. ([article](#), [arXiv](#))
Invited contribution to special volume honoring Anthony M. Bloch.
- 6 **Planning of obstacle-aided navigation for multi-legged robots using a sampling-based method over directed graphs**
Kaustav Chakraborty, Haodi Hu, Matthew D. Kvalheim, and Feifei Qian
IEEE Robotics and Automation Letters, 7.4 (2022), pp. 8861–8868. ([article](#))
- 5 **Families of periodic orbits: closed 1-forms and global continuability**
Matthew D. Kvalheim and Anthony M. Bloch
J. Differential Equations, 285 (2021), pp. 211–257. ([article](#), [arXiv](#))

- 4 **Existence and uniqueness of global Koopman eigenfunctions for stable fixed points and periodic orbits**
Matthew D. Kvalheim and Shai Revzen
 Physica D, 425 (2021), 132959. ([article](#), [arXiv](#))
- 3 **Conley's fundamental theorem for a class of hybrid systems**
Matthew D. Kvalheim, Paul Gustafson, and Daniel E. Koditschek
 SIAM J. Applied Dynamical Systems, 20.2 (2021), pp. 784–825. ([article](#), [arXiv](#))
- 2 **Gait modeling and optimization for the perturbed Stokes regime**
Matthew D. Kvalheim, Brian Bittner, and Shai Revzen
 Nonlinear Dynamics, 97.4 (2019), pp. 2249–2270. ([article](#), [arXiv](#))
- 1 **Global linearization and fiber bundle structure of invariant manifolds**
Jaap Eldering, Matthew D. Kvalheim, and Shai Revzen
 Nonlinearity, 31.9 (2018), pp. 4202–4245. ([article](#), [arXiv](#))

Peer-Reviewed Conference Publications

- 4 **Relationships between necessary conditions for feedback stabilizability**
Matthew D. Kvalheim
 Geometry, Topology, and Control System Design: Proceedings of a Banff International Research Station Workshop (2025), pp. 167–179. ([book](#), [arXiv](#))
- 3 **The role of symmetry in constructing geometric flat outputs for free-flying robotic systems**
Jake Welde, Matthew D. Kvalheim, and Vijay Kumar
 2023 IEEE International Conference on Robotics and Automation (2023), pp. 12247–12253. ([article](#), [arXiv](#))
- 2 **Generic properties of Koopman eigenfunctions for stable fixed points and periodic orbits**
Matthew D. Kvalheim, David A. Hong, and Shai Revzen
 IFAC-PapersOnline, 54.9 (2021), pp. 267–272. ([article](#), [arXiv](#))
- 1 **Data-driven models of legged locomotion**
Shai Revzen and Matthew D. Kvalheim
 Micro-and Nanotechnology Sensors, Systems, and Applications VII, 9467 (2015). ([article](#))

Book Chapters

- 2 **Bioinspired legged locomotion, chapter 3.2: Templates and anchors**
Matthew D. Kvalheim and Shai Revzen
 Butterworth-Heinemann, Elsevier, Oxford (2017)
- 1 **Bioinspired legged locomotion, chapter 3.5: Locomotion as an oscillator**
Shai Revzen and Matthew D. Kvalheim
 Butterworth-Heinemann, Elsevier, Oxford (2017)

Preprints

- 6 **The Maxwell conjecture is false**
Philip Arathoon, Gavin Ball, and Matthew D. Kvalheim
 Under review in J. European Mathematical Society (2026). ([arXiv](#))
- 5 **Asymptotic stability and differential topology**
Matthew D. Kvalheim
 Under review in Analysis and Applications (2026). ([preprint](#))
 Invited contribution to special volume honoring Stephen Smale.

- 4 **Stabilizability of first-order dynamics in second-order systems**
Matthew D. Kvalheim
 Under review in SIAM J. Control and Optimization (2026). ([arXiv](#))
- 3 **Differential topology of the spaces of asymptotically stable vector fields and Lyapunov functions**
Matthew D. Kvalheim
 To be submitted (2025). ([arXiv](#))
- 2 **Autoencoding dynamics: topological limitations and capabilities**
Matthew D. Kvalheim and Eduardo D. Sontag
 Working paper (2025). ([arXiv](#))
- 1 **Koopman embedding and super-linearization counterexamples with isolated equilibria**
Philip Arathoon and Matthew D. Kvalheim
 Working paper (2023). ([arXiv](#))

Invited Talks

- 2026 **On the stabilizability of reference dynamics for mechanical systems**, *65th IEEE Conference on Decision and Control (CDC)*, Honolulu, HI
 Minisymposium: Geometric Control and Novel Applications
- 2026 **TBD**, *Workshop: Koopman Operator Theory: Fundamentals, Approximations, and Applications 3*, Kyoto, Japan
- 2026 **TBD**, *University of Delaware*, Newark, DE, USA
 Robotics GSO-CARS Seminar Series
- 2026 **TBD**, *University of California, Santa Cruz*, Santa Cruz, CA, USA
 Mathematics Department Colloquium
- 2026 **On the topology of asymptotic stability**, *George Mason University*, Arlington, VA, USA
 2026 AFOSR Dynamical Systems and Control Theory Review
- 2026 **Topological limitations and capabilities for autoencoding dynamics**, *SIAM Annual Meeting*, Cleveland, OH
 Minisymposium: Topology, Geometry and Physics in High-Dimensional Representation Learning
- 2026 **On the universality of linear dynamics**, *International Society in Quantization, Geometry, and Dynamics*, online
 Session: Ergodic Theory and Topological Dynamics
- 2026 **On the universality of linear dynamics and Koopman embeddings**, *Georgetown University*, Washington, DC (moved online due to last-minute illness)
 Department of Mathematics and Statistics Colloquium Series
- 2025 **When do practical Koopman embeddings exist?**, *61st Annual Allerton Conference*, University of Illinois, Urbana-Champaign, USA
 Session: Dynamical Systems, Learning, and Control
- 2025 **On the (non)existence of Koopman embeddings**, *Basic Research Innovation Collaboration Center*, Arlington, VA, USA
 2025 AFOSR Dynamical Systems and Control Theory Review
- 2025 **When do Koopman embeddings exist?**, *Workshop: Koopman Operator Theory: Fundamentals, Approximations, and Applications 2*, Cres, Croatia
- 2025 **On Professor Smale's legacy for asymptotic stability theory**, *Smale@95: A Conference in Honor of Steve Smale*, Simons Institute, Berkeley, CA, USA ([link](#))

- 2025 **When do Koopman embeddings exist?**, *SIAM Conference on Applications of Dynamical Systems*, Denver, CO, USA
Minisymposium: Theory of Koopman Operators for Data Driven Dynamical Systems
- 2025 **On the capabilities and limitations of autoencoders**, *Joint Mathematics Meetings*, Seattle, WA, USA
with Eduardo D. Sontag
Special session: Control theory and artificial intelligence
- 2024 **Obstructions to feedback stabilization**, *Basic Research Innovation Collaboration Center*, Arlington, VA, USA
2024 AFOSR Dynamical Systems and Control Theory Review
- 2024 **Identifying engineering (im)possibilities with geometry and topology**, *University of Delaware*, Newark, DE, USA
Electrical and Computer Engineering Spring Seminar Series
- 2024 **Linearizability of dynamical systems by embeddings**, *Joint Mathematics Meetings*, San Francisco, CA, USA
Minisymposium: Geometry and Symmetry in Differential Equations, Control, and Applications
- 2023 **Discovering engineering (im)possibilities with geometry and topology**, *Johns Hopkins University Applied Physics Laboratory*, Baltimore, MD, USA
- 2023 **Discovering engineering (im)possibilities with geometry and topology**, *University of Pennsylvania*, Philadelphia, PA, USA
GRASP SFI Seminar ([link](#))
- 2023 **Asymptotic stabilizability**, *Banff International Research Station*, Banff, AB, Canada
Workshop: Geometry, Topology and Control System Design ([link](#))
- 2022 **Existence and uniqueness of Koopman eigenfunctions near stable equilibria and limit cycles**, *International Symposium on Nonlinear Theory and Its Applications*, online
- 2022 **Negative resistance in small-noise dynamics via persistent homology**, *Indian Institute of Technology Bombay*, online
SysConTalks
- 2022 **Nonlinear systems**, *University of Michigan*, Ann Arbor, MI
- 2022 **Large deviations, persistent homology, and Brownian conductors with negative resistance**, *Cornell University*, online
Probability Seminar
- 2022 **When can hybrid systems operate safely?**, *University of Michigan*, online
Control Seminar
- 2022 **Flux in small noise dynamics: persistence and negative resistance**, *Brown University, Division of Applied Mathematics*, online
- 2021 **Flux in small noise dynamics: negative resistance and persistence**, *University of Illinois Urbana-Champaign*, online
Mathematical Biology Seminar
- 2021 **When and how are hybrid dynamical systems conjugate to their classical quotients?**, *SIAM Conference on Control and its Applications*, online
with Samuel A. Burden
Minisymposium: Physically grounded semantics for programming hybrid dynamical systems

- 2021 **Toward a physically grounded type theory for robot task composition**, *IEEE International Conference on Robotics and Automation*, online
with Daniel E. Koditschek and Paul Gustafson
Workshop: Compositional robotics: mathematics and tools
- 2021 **Toward a physically grounded type theory for robot task composition**, *IEEE International Conference on Robotics and Automation*, online
with Daniel E. Koditschek and Paul Gustafson
Workshop: Compositional robotics: mathematics and tools
- 2021 **Small noise dynamics, persistence, and negative resistance**, *SIAM Conference on Applications of Dynamical Systems*, online
with Yuliy Baryshnikov
Minisymposium: Stochastic oscillators
- 2021 **Flux in tilted potential systems: negative resistance and persistence**, *Army Research Office SLICE MURI W911NF-18-1-032 Group Meeting*, online
- 2021 **Towards a Conley theory for hybrid dynamical systems**, *Rutgers University*, online
TRIPODS/DATA-INSPIRE Workshop: Dynamics, Topology, and Robotic Control
- 2021 **Hierarchical composition via collapse of dimension in dynamical systems**, *ETH Zürich*, online
Series: Autonomy Talks
- 2021 **Existence and uniqueness of global linearizing coordinates**, *Army Research Lab*, online
Representing the MURI group funded by Army Research Office MURI 911NF-17-1-0306
- 2021 **Safety and stabilization: necessary conditions**, *Army Research Office SLICE MURI W911NF-18-1-032 Group Meeting*, online
- 2021 **Necessary conditions for feedback stabilization and safety**, *University of Pennsylvania*, online
Kod*Lab Research Seminar
- 2020 **Proving periodic orbits exist: global continuation and Lyapunov 1-forms**, *Invited Talk, University of Pennsylvania*, online
Graduate Research Seminar in Applied Topology
- 2020 **Proving periodic orbits exist: Lyapunov 1-forms and global continuation**, *University of Pennsylvania*, online
Kod*Lab Research Seminar
- 2020 **Geometry and dynamics of circulant systems**, *AMS sectional meeting*, online
with Anthony M. Bloch (speaker)
Special session: Geometric dynamics
- 2020 **Conley's fundamental theorem for a class of hybrid systems**, *Army Research Office SLICE MURI W911NF-18-1-032 Group Meeting*, online
- 2020 **Conley's fundamental theorem for a class of hybrid systems**, *University of Pennsylvania*, online
Kod*Lab Research Seminar
- 2020 **Toward a task planning theory for robot hybrid dynamics**, *Casa Matemática Oaxaca*, online
Workshop: Topological complexity and motion planning
- 2020 **Isochrons from short, noisy data**, *Dynamics Days Europe*, online
with Simon Wilshin, Clayton Scott, and Shai Revzen
Minisymposium: Stochastic oscillators

- 2019 **Invariant manifolds, locomotion, and the Koopman operator**, *Invited Talk, University of Pennsylvania*, Philadelphia, PA
Chaudhari Research Group Seminar
- 2019 **Templates, anchors, and normally hyperbolic invariant manifolds**, *Invited Talk, University of Pennsylvania*, Philadelphia, PA
DAIR Lab Research Seminar
- 2019 **Templates, anchors, and normally hyperbolic invariant manifolds**, *Invited Talk, University of Pennsylvania*, Philadelphia, PA
ScalAR Lab Research Seminar
- 2019 **Existence and uniqueness of linearizing semiconjugacies for stable fixed points and periodic orbits**, *California Institute of Technology*, Pasadena, CA
Army Research Office MURI 911NF-17-1-0306 Group Meeting
- 2019 **Dynamics of circulant systems of ODEs**, *SIAM Conference on Applications of Dynamical Systems*, Snowbird, UT
with Anthony M. Bloch (*speaker*)
- 2019 **Reduced-order models for locomotion in the perturbed Stokes regime**, *SIAM Conference on Applications of Dynamical Systems*, Snowbird, UT
with Shai Revzen
- 2019 **Hybrid oscillators: phase and amplitude in a class of non-smooth systems**, *SIAM Conference on Applications of Dynamical Systems*, Snowbird, UT
with Shai Revzen
- 2018 **Aspects of invariant manifold theory and applications**, *University of California Santa Barbara*, Santa Barbara, CA
Army Research Office MURI 911NF-17-1-0306 Group Meeting
- 2018 **Oscillators, asymptotic phase, and reduction of dynamical systems**, *University of Michigan*, Ann Arbor, MI
Applied and Interdisciplinary Mathematics Student Seminar
- 2018 **Invariant manifolds, asymptotic phase, and data-driven algorithms**, *University of Pennsylvania*, Philadelphia, PA
Kod*Lab Research Seminar
- 2016 **A recipe for embedding templates in anchors**, *University of Michigan*, Ann Arbor, MI
Legged Robotics Seminar
- 2015 **Introduction to homology: towards topological data analysis, pt. II**, *University of Michigan*, Ann Arbor, MI
Signal Processing Student Seminar
- 2015 **Introduction to homology: towards topological data analysis, pt. I/II**, *University of Michigan*, Ann Arbor, MI
Signal Processing Student Seminar

Other Talks

- 2023 **A compositional approach to certifying the almost global asymptotic stability of cascade systems**, *IEEE International Conference on Robotics and Automation*, London, UK
with Jake Welde (*speaker*) and Vijay Kumar
Workshop: Compositional robotics: mathematics and tools

- 2023 **A principal bundle perspective on differential flatness in complex robotic and biological systems**, *Americal Physical Society March Meeting*, Las Vegas, Nevada, USA
with Jake Welde (speaker) and Vijay Kumar
- 2019 **Is legged locomotion almost smooth?**, *Dynamic Walking*, Calgary, AB, Canada
with Shai Revzen (speaker) and George Council
- 2018 **Testing an extended ‘Posture Principle’**, *Annual meeting of the Society for Integrative and Comparative Biology*, San Francisco, CA
with Shai Revzen
- 2015 **Phase-based models of rhythmic systems**, *Dynamic Walking*, Columbus, OH
with Shai Revzen
- 2015 **Better models of rhythmic systems: predicting locomotion from phase alone**, *Annual meeting of the Society for Integrative and Comparative Biology*, West Palm Beach, FL
with Shai Revzen
- 2014 **Why the trot?**, *Annual meeting of the Society for Integrative and Comparative Biology*, Austin, TX
with Shai Revzen and Sam Burden (speakers)

Posters

- 2023 **The role of symmetry in constructing geometric flat outputs for free-flying robotic systems**, *IEEE International Conference on Robotics and Automation*, London, UK
with Jake Welde (presenter) and Vijay Kumar
- 2020 **A mode map representation to predict steady states and attraction basins for legged locomotion on obstacle terrains**, *International Conference on Intelligent Robotics and Systems (IROS)*, online
with Haodi Hu (presenter), Michelle Joyce, Simon Wilshin, Andrew Spence, and Feifei Qian
Workshop: Robotics-inspired biology
- 2019 **Principal Koopman eigenfunctions for nonlinear and nonsmooth systems**, *University of California, Los Angeles*, Los Angeles, CA
with Shai Revzen
Workshop: Operator theoretic methods in dynamic data analysis and control
- 2018 **Global linearization and fiber bundle structure of invariant manifolds**, *AIMS Conference Series on Dynamical Systems and Differential Equations*, Taipei, Taiwan
with Jaap Eldering and Shai Revzen
- 2018 **Templates and Anchors: a review of notions of model reduction**, *Dynamic Walking*, Pensicola, FL
with Shai Revzen
- 2017 **Asymptotic phase, model reduction, and control of Templates & Anchors**, *Ohio State University, Mathematical Biosciences Institute*, Columbus, OH
with Shai Revzen
Workshop: Sensorimotor control of animals and robots
- 2017 **A dynamical systems perspective on Templates & Anchors: some general methods for anchoring templates**, *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, Vancouver, BC
with Shai Revzen
Workshop: Robotics-inspired Biology

- 2014 **Phase-based algorithm for modeling time-rescaling distortion of rhythmic data**, *University of Michigan*, Ann Arbor, MI
with Shai Revzen
Engineering Graduate Symposium
- 2013 **A hybrid system provides a robust alternative to a linear regulator**, *University of Michigan*, Ann Arbor, MI
with Shai Revzen
Engineering Graduate Symposium
- 2013 **An indoor navigation device for use in GPS-denied environments**, *Ohio University*, Athens, OH, Award: 1st place, EECS division
with Ryan Kollar, Christopher Moore, Jessica Belzer, and Matthew Miltner
Ohio University Student Expo

Academic Service

Editorial Activity

- 2026 **Invited Organizer and Editor**, *Special Collection: Mathematical Foundations of Feedback Control*, *Mathematics of Control, Signals, and Systems*
- 2025 **Associate Editor and International Program Committee Member**, *IFAC Conference on Networked Systems (NecSys2025)*
- 2023– **Associate Editor**, *Mathematics of Control, Signals, and Systems*

Other Professional Service

- 2023 **Co-organizer**, *IEEE International Conference on Robotics and Automation*
Workshop: “Compositional Robotics: Mathematics and Tools”
- 2022 **Co-organizer**, *Festschrift in honor of Daniel E. Koditschek* ([link](#))
- 2022 **Co-organizer**, *IEEE International Conference on Robotics and Automation*
Workshop: “Compositional Robotics: Mathematics and Tools”

Reviewing for funding agencies, *AFOSR Dynamical Systems and Control Theory Program*; *NSF Division of Mathematical Sciences Computational and Data-Enabled Science and Engineering Program*

Reviewing and quick opinions for journals, *Annual Reviews in Control*; *ASME J. Biomechanical Engineering*; *Automatica*; *Biological Cybernetics*; *Chaos*; *IEEE Control Systems Letters*; *IEEE Transactions on Automatic Control*; *J. London Mathematical Society*; *J. Machine Learning Research*; *Machine Learning: Science and Technology*; *Mathematics of Control, Signals, and Systems*; *Science Advances*; *Scientific Reports*; *SIAM J. Control and Optimization*; *Stochastic Models*

Reviewing for conferences, *European Control Conference*; *IEEE Conference on Decision and Control*; *IEEE International Conference on Robotics and Automation*; *IFAC World Congress*; *International Symposium on Nonlinear Theory and Its Applications*; *Learning for Dynamics & Control*; *Robotics: Science and Systems*

Member, *American Mathematical Society*, *Institute of Electrical and Electronics Engineers*, *Society for Industrial and Applied Mathematics*, *SIAM Activity Groups on Dynamical Systems and on Control and Systems Theory*, *Pi Mu Epsilon*, *Tau Beta Pi*

University Service

- 2026 **Volunteer**, *UMBC Invitational Science Olympiad*, *University of Maryland*, Baltimore County, MD

- 2024–2025 **Co-organizer and volunteer**, *UMBC Invitational Science Olympiad*, University of Maryland, Baltimore County, MD
- 2024 **Volunteer, Faculty Meet-and-Greet, Admitted Student Sneak Peek**, *College of Natural and Mathematical Sciences*, University of Maryland, Baltimore County, MD
- 2023–2024 **Co-organizer and volunteer**, *UMBC Invitational Science Olympiad*, University of Maryland, Baltimore County, MD
- 2022 **Volunteer**, *Females Excelling More in Math, Engineering, and Science (F.E.M.M.E.S.)*, University of Michigan, Ann Arbor, MI
- [Departmental Service](#)
- 2026 **Committee member**, *Departmental PhD oral exams*, University of Maryland, Baltimore County, MD
- 2024–2026 **Co-author and grader**, *Departmental PhD comprehensive exams*, University of Maryland, Baltimore County, MD
- 2024–2026 **Adviser, Pi Mu Epsilon Chapter**, *University of Maryland, Baltimore County, MD* Mathematics honor society
- 2023–2024 **Co-adviser, Pi Mu Epsilon Chapter**, *University of Maryland, Baltimore County, MD* Mathematics honor society
- 2019 **Volunteer, demonstrations to visiting K-12 students, Philly Robotics Expo (PRX)**, *University of Pennsylvania*, Philadelphia, PA
Robotics outreach
- 2019 **Volunteer, demonstrations to members of the National Society of Black Engineers**, *Detroit, MI*, Representing BIRDS Laboratory, University of Michigan; robotics outreach
45th Annual Convention of the National Society of Black Engineers (NSBE)
- 2015 **Volunteer, demonstrations to alumni and industry**, *Engineering Graduate Symposium*, Ann Arbor, MI
Representing BIRDS Laboratory, University of Michigan; robotics outreach
- 2015 **Volunteer, panel participant**, *Graduate Student Council*, Ann Arbor, MI
Member of panel of University of Michigan graduate students; answering questions from prospective graduate students
- 2014 **Volunteer, demonstrations to elementary schoolers**, *Bryant Elementary School*, Ann Arbor, MI
Representing BIRDS Laboratory, University of Michigan; robotics outreach

Teaching Experience

- 2026 **Course Instructor**, *University of Maryland, Baltimore County, MD*
Instructor of record for reading course on Smooth Manifolds (Math 699) in Fall 2026
- 2026 **Course Instructor**, *University of Maryland, Baltimore County, MD*
Instructor of record for Continuous Dynamical Systems (Math 487) in Fall 2026
- 2026 **Course Instructor**, *University of Maryland, Baltimore County, MD*
Instructor of record for Introduction to Differential Topology (Math 423/673) in Spring 2026
- 2025 **Course Instructor**, *University of Maryland, Baltimore County, MD*
Instructor of record for Continuous Dynamical Systems (Math 487) in Fall 2025
- 2025 **Course Instructor**, *University of Maryland, Baltimore County, MD*
Instructor of record for reading course on Differential Topology (Math 699) in Fall 2024

- 2025 **Course Instructor**, *University of Maryland, Baltimore County*, MD
Instructor of record for Ordinary Differential Equations (Math 612) in Spring 2025
- 2024 **Course Instructor**, *University of Maryland, Baltimore County*, MD
Instructor of record for reading course on Smooth Manifolds (Math 499 and 699) in Fall 2024
- 2024 **Course Instructor**, *University of Maryland, Baltimore County*, MD
Instructor of record for Real Analysis (Math 600) in Fall 2024
- 2024 **Course Instructor**, *University of Maryland, Baltimore County*, MD
Instructor of record for Ordinary Differential Equations (Math 612) in Spring 2024
- 2023 **Course Instructor**, *University of Maryland, Baltimore County*, MD
Instructor of record for Introduction to Linear Algebra (Math 221) in Fall 2023
- 2023 **Course Instructor**, *University of Michigan*, Ann Arbor, MI
Instructor of record for two course sections of Differential Equations (Math 316) in Winter 2023
- 2022 **Course Instructor**, *University of Michigan*, Ann Arbor, MI
Instructor of record for two course sections of Calculus I (Math 115) in Fall 2022
- 2019 **Guest Lecturer**, *University of Pennsylvania*, Philadelphia, PA
Graduate course: Dynamical Systems for Engineering and Biological Applications (ESE 512)
Course instructor: Daniel E. Koditschek
- 2014 **Graduate Student Instructor**, *University of Michigan*, Ann Arbor, MI
Senior-level course: Discrete-Time Signal Processing (EECS 451)
Course instructor: Laura Balzano
- 2011–2012 **Supplemental Instruction Leader**, **Academic Achievement Center**, *Ohio University*, Athens, OH
Sophomore-level course: Statics (CE 220)

Mentoring

- 2025– **Advisor of graduate student**, *University of Maryland*, Baltimore County, MD
Advising 1 graduate student in research
- 2022–2023 **Mentor of graduate student**, *University of Michigan*, Ann Arbor, MI
Mentored 1 graduate student in research
- 2019–2023 **Mentor of graduate students**, *University of Pennsylvania*, Philadelphia, PA
Mentored 2 graduate students in research
- 2013–2018 **Mentor of undergraduate and graduate students**, **BIRDS Laboratory**, *University of Michigan*
Mentored 2 undergraduate and 3 graduate students

Trainings and Certifications

- 2025 **Minicourse on rational homotopy of diffeomorphisms of discs**, *Kansas State University*, Manhattan, KS
Instructor: Alexander Kupers
- 2024 **Madrid Summer School on h-principle**, *Instituto de Ciencias Matemáticas*, Madrid, Spain
Instructors: M Bertelson, K Cieliebak, Y Eliashberg, A Kupers, O Lazarev, G Meigniez, R Cardona, D Álvarez-Gavela, Á del Pino Gómez, M Theillièrre
- 2024 **Spring Faculty Success Program**, *National Center for Faculty Development and Diversity (NCFDD)*, online
Nominated by the Office of the Provost at the University of Maryland, Baltimore County

- 2023 **Principal Investigator Research Administrative Training & Education**, *Office of the Vice President for Research, UMBC, Baltimore, MD*
- 2023 **Media and Communications Training for STEM Faculty**, *American Association for the Advancement of Sciences (AAAS), Baltimore, MD*

Media Activities

- 2025 **UMBC mathematician honored with invitation to Stephen Smale's 95th birthday conference**
Author: Sarah Hansen ([article](#))
- 2025 **In the world of math, the hunt for eloquent solutions excites these researchers**
Author: Sarah Hansen ([article](#))